CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 13: Statistics

Step-by-step solutions to the textbook exercises: 22 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 13.1

Q1 The numbers of plants in 20 houses are given below. Find the mean number of plants per house. Which method did you use, and why?

Number of plants	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12	12 – 14
Number of houses	1	2	1	5	6	2	3

SOLUTION

The class marks and frequencies are small numbers, so the direct method is the easiest.

Class mark x_i	1	3	5	7	9	11	13	Total
f_i	1	2	1	5	6	2	3	20
$f_i x_i$	1	6	5	35	54	22	39	162

Mean $= \dfrac{\sum f_i x_i}{\sum f_i} = \dfrac{162}{20} = 8.1.$

Answer: 8.1 plants per house, by the direct method

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Q2 The daily wages of 50 workers of a factory are given below. Find the mean daily wage by a suitable method.

Daily wages (₹)	500 – 520	520 – 540	540 – 560	560 – 580	580 – 600
Number of workers	12	14	8	6	10

SOLUTION

Use the step-deviation method with $a = 550$, $h = 20$ and $u_i = \dfrac{x_i - a}{h}$.

x_i	510	530	550	570	590	Total
f_i	12	14	8	6	10	50
u_i	-2	-1	0	1	2	
$f_i u_i$	-24	-14	0	6	20	-12

$$\text{Mean} = a + \frac{\sum f_i u_i}{\sum f_i} \times h = 550 + \frac{-12}{50} \times 20 = 550 - 4.8 = 545.2.$$

Answer: ₹545.20

Q3 The daily pocket allowance of the children of a locality is given below. The mean is ₹18. Find the missing frequency f.

Allowance (₹)	11 – 13	13 – 15	15 – 17	17 – 19	19 – 21	21 – 23	23 – 25
Number of children	7	6	9	13	f	5	4

SOLUTION

The class marks are 12, 14, 16, 18, 20, 22, 24.

$\sum f_i = 44 + f$ and $\sum f_i x_i = 84 + 84 + 144 + 234 + 20f + 110 + 96 = 752 + 20f$.

$\dfrac{752 + 20f}{44 + f} = 18$ gives $752 + 20f = 792 + 18f$, so $2f = 40$.

Answer: $f = 20$

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Q4 The heartbeats per minute of 30 women are summarised below. Find the mean by a suitable method.

Heartbeats per minute	65 – 68	68 – 71	71 – 74	74 – 77	77 – 80	80 – 83	83 – 86
Number of women	2	4	3	8	7	4	2

SOLUTION

Use the step-deviation method with $a = 75.5$ and $h = 3$.

x_i	66.5	69.5	72.5	75.5	78.5	81.5	84.5	Total
f_i	2	4	3	8	7	4	2	30
u_i	-3	-2	-1	0	1	2	3	
f_iu_i	-6	-8	-3	0	7	8	6	4

Mean $= 75.5 + \dfrac{4}{30} \times 3 = 75.5 + 0.4 = 75.9.$

Answer: 75.9 heartbeats per minute

Q5 Boxes of mangoes hold different numbers of mangoes, as shown below. Find the mean number of mangoes in a box. Which method did you choose?

Number of mangoes	50 – 52	53 – 55	56 – 58	59 – 61	62 – 64
Number of boxes	15	110	135	115	25

SOLUTION

The class marks are 51, 54, 57, 60, 63. (They are the same whether or not the classes are made continuous.) Use the step-deviation method with $a = 57$ and $h = 3$.

x_i	51	54	57	60	63	Total
f_i	15	110	135	115	25	400
u_i	-2	-1	0	1	2	
f_iu_i	-30	-110	0	115	50	25

Mean $= 57 + \dfrac{25}{400} \times 3 = 57 + 0.1875 \approx 57.19.$

Answer: about 57.19 mangoes, by the step-deviation method

Q6 The daily expenditure on food of 25 households is given below. Find the mean by a suitable method.

Daily expenditure (₹)	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
Number of households	4	5	12	2	2

SOLUTION

Use the step-deviation method with $a = 225$ and $h = 50$.

x_i	125	175	225	275	325	Total
f_i	4	5	12	2	2	25
u_i	-2	-1	0	1	2	
$f_i u_i$	-8	-5	0	2	4	-7

$$\text{Mean} = 225 + \frac{-7}{25} \times 50 = 225 - 14 = 211.$$

Answer: ₹211

Q7 The concentration of SO_2 in the air (in ppm) in 30 localities is given below. Find the mean concentration.

Concentration (ppm)	0.00 – 0.04	0.04 – 0.08	0.08 – 0.12	0.12 – 0.16	0.16 – 0.20	0.20 – 0.24
Frequency	4	9	9	2	4	2
x_i	0.02	0.06	0.10	0.14	0.18	0.22
$f_i x_i$	0.08	0.54	0.90	0.28	0.72	0.44

SOLUTION

$\sum f_i = 30$ and $\sum f_i x_i = 2.96$.

$$\text{Mean} = \frac{2.96}{30} \approx 0.099.$$

Answer: about 0.099 ppm

Q8 The absentee record of 40 students for a term is given below. Find the mean number of days a student was absent.

Number of days	0 – 6	6 – 10	10 – 14	14 – 20	20 – 28	28 – 38	38 – 40
Number of students	11	10	7	4	4	3	1

SOLUTION

The class widths are unequal, so use the direct method.

x_i	3	8	12	17	24	33	39	Total
$f_i x_i$	33	80	84	68	96	99	39	499

Mean $= \dfrac{499}{40} = 12.475 \approx 12.48.$

Answer: about 12.48 days

Q9 The literacy rates (in %) of 35 cities are given below. Find the mean literacy rate.

Literacy rate (%)	45 – 55	55 – 65	65 – 75	75 – 85	85 – 95
Number of cities	3	10	11	8	3

SOLUTION

Use the step-deviation method with $a = 70$ and $h = 10$.

x_i	50	60	70	80	90	Total
u_i	-2	-1	0	1	2	
$f_i u_i$	-6	-10	0	8	6	-2

Mean $= 70 + \dfrac{-2}{35} \times 10 = 70 - 0.57 = 69.43.$

Answer: about 69.43%

2 Exercise 13.2

Q1 The ages of patients admitted to a hospital in a year are given below. Find the mode and the mean, and compare them.

Age (years)	5 – 15	15 – 25	25 – 35	35 – 45	45 – 55	55 – 65
Number of patients	6	11	21	23	14	5

SOLUTION

The modal class is 35 – 45, with $l = 35$, $h = 10$, $f_1 = 23$, $f_0 = 21$, $f_2 = 14$.

Mode $= l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 35 + \dfrac{2}{46 - 35} \times 10 = 35 + 1.82 = 36.8$ years.

For the mean, the class marks are 10, 20, 30, 40, 50, 60: $\sum f_i x_i = 60 + 220 + 630 + 920 + 700 + 300 = 2830$ and $\sum f_i = 80$. Mean $= \dfrac{2830}{80} = 35.37$ years.

The largest number of patients are about 36.8 years old, while the average age of a patient is about 35.37 years.

Answer: mode ≈ 36.8 years, mean ≈ 35.37 years

Q2 The lifetimes (in hours) of 225 electrical components are given below. Find the modal lifetime.

Lifetime (hours)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120
Frequency	10	35	52	61	38	29

SOLUTION

The modal class is 60 – 80, with $l = 60$, $h = 20$, $f_1 = 61$, $f_0 = 52$, $f_2 = 38$.

Mode $= 60 + \dfrac{61 - 52}{122 - 52 - 38} \times 20 = 60 + \dfrac{9}{32} \times 20 = 65.625$.

Answer: 65.625 hours

Q3 The monthly household expenditure of 200 families is given below. Find the modal and the mean monthly expenditure.

Expenditure (₹)	1000 – 1500	1500 – 2000	2000 – 2500	2500 – 3000	3000 – 3500	3500 – 4000	4000 – 4500	4500 – 5000
Number of families	24	40	33	28	30	22	16	7

SOLUTION

The modal class is 1500 – 2000, with $l = 1500$, $h = 500$, $f_1 = 40$, $f_0 = 24$, $f_2 = 33$.

Mode $= 1500 + \dfrac{40 - 24}{80 - 24 - 33} \times 500 = 1500 + \dfrac{16}{23} \times 500 \approx 1847.83.$

For the mean use $a = 2750$, $h = 500$. The values of u_i are $-3, -2, -1, 0, 1, 2, 3, 4$, so $\sum f_i u_i = -72 - 80 - 33 + 0 + 30 + 44 + 48 + 28 = -35.$

Mean $= 2750 + \dfrac{-35}{200} \times 500 = 2750 - 87.5 = 2662.5.$

Answer: mode ≈ ₹1847.83, mean = ₹2662.50

Q4 The state-wise teacher-student ratio in higher secondary schools is given below. Find the mode and the mean, and interpret them.

Students per teacher	15 – 20	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55
Number of states / U.T.	3	8	9	10	3	0	0	2

SOLUTION

The modal class is 30 – 35, with $l = 30$, $h = 5$, $f_1 = 10$, $f_0 = 9$, $f_2 = 3$.

Mode $= 30 + \dfrac{10 - 9}{20 - 9 - 3} \times 5 = 30 + 0.625 \approx 30.6.$

For the mean, the class marks are 17.5, 22.5, ..., 52.5: $\sum f_i x_i = 52.5 + 180 + 247.5 + 325 + 112.5 + 0 + 0 + 105 = 1022.5$ and $\sum f_i = 35$. Mean $= \dfrac{1022.5}{35} \approx 29.2.$

Most states and U.T.s have about 30.6 students per teacher, and on average there are about 29.2 students per teacher.

Answer: mode ≈ 30.6, mean ≈ 29.2

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Q5 The runs scored by some top batsmen in one-day internationals are given below. Find the mode.

Runs scored	3000 – 4000	4000 – 5000	5000 – 6000	6000 – 7000	7000 – 8000	8000 – 9000	9000 – 10000	10000 – 11000
Number of batsmen	4	18	9	7	6	3	1	1

SOLUTION

The modal class is 4000 – 5000, with $l = 4000$, $h = 1000$, $f_1 = 18$, $f_0 = 4$, $f_2 = 9$.

$$\text{Mode} = 4000 + \frac{18 - 4}{36 - 4 - 9} \times 1000 = 4000 + \frac{14}{23} \times 1000 \approx 4608.7.$$

Answer: about 4608.7 runs

Q6 The number of cars passing a spot was noted for 100 periods of 3 minutes each, as below. Find the mode.

Number of cars	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Frequency	7	14	13	12	20	11	15	8

SOLUTION

The modal class is 40 – 50, with $l = 40$, $h = 10$, $f_1 = 20$, $f_0 = 12$, $f_2 = 11$.

$$\text{Mode} = 40 + \frac{20 - 12}{40 - 12 - 11} \times 10 = 40 + \frac{80}{17} \approx 44.7.$$

Answer: about 44.7 cars

MathSetu

3 Exercise 13.3

Q1 The monthly electricity consumption of 68 consumers is given below. Find the median, mean and mode, and compare them.

Consumption (units)	65 – 85	85 – 105	105 – 125	125 – 145	145 – 165	165 – 185	185 – 205
Number of consumers	4	5	13	20	14	8	4
Cumulative frequency	4	9	22	42	56	64	68

SOLUTION

Median: $\frac{n}{2} = 34$, so the median class is 125 – 145, with $l = 125$, $cf = 22$, $f = 20$, $h = 20$. Median $= l + \frac{\frac{n}{2} - cf}{f} \times h = 125 + \frac{34 - 22}{20} \times 20 = 137$.

Mode: the modal class is 125 – 145, with $f_1 = 20$, $f_0 = 13$, $f_2 = 14$. Mode $= 125 + \frac{7}{40 - 27} \times 20 \approx 135.76$.

Mean: with $a = 135$, $h = 20$ and $u_i = -3, -2, -1, 0, 1, 2, 3$, $\sum f_i u_i = -12 - 10 - 13 + 0 + 14 + 16 + 12 = 7$. Mean $= 135 + \frac{7}{68} \times 20 \approx 137.06$.

The three measures are close to one another, at about 137 units.

Answer: median 137 units, mean ≈ 137.06 units, mode ≈ 135.76 units

Q2 The median of the distribution below is 28.5 and the total frequency is 60. Find x and y.

Class interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	5	x	20	15	y	5

SOLUTION

$5 + x + 20 + 15 + y + 5 = 60$ gives $x + y = 15$.

The median 28.5 lies in 20 – 30, so $l = 20$, $f = 20$, $h = 10$, $cf = 5 + x$ and $\frac{n}{2} = 30$.

$28.5 = 20 + \frac{30 - (5 + x)}{20} \times 10$ gives $8.5 = \frac{25 - x}{2}$, so $x = 8$ and $y = 7$.

Answer: $x = 8$, $y = 7$

Q3 The ages of 100 policy holders are given as a "less than" distribution: below 20: 2, below 25: 6, below 30: 24, below 35: 45, below 40: 78, below 45: 89, below 50: 92, below 55: 98, below 60: 100. Policies are given only to persons aged 18 or more but less than 60. Find the median age.

SOLUTION

Subtract each cumulative frequency from the next to get the class frequencies.

Age (years)	18 – 20	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55	55 – 60
Frequency	2	4	18	21	33	11	3	6	2
Cumulative frequency	2	6	24	45	78	89	92	98	100

$\frac{n}{2} = 50$, so the median class is 35 – 40, with $l = 35$, $cf = 45$, $f = 33$, $h = 5$.

Median $= 35 + \frac{50 - 45}{33} \times 5 \approx 35.76$.

Answer: about 35.76 years

Q4 The lengths of 40 leaves, measured to the nearest millimetre, are given below. Find the median length.

Length (mm)	118 – 126	127 – 135	136 – 144	145 – 153	154 – 162	163 – 171	172 – 180
Number of leaves	3	5	9	12	5	4	2
Cumulative frequency	3	8	17	29	34	38	40

SOLUTION

The classes are not continuous, so change them to 117.5 – 126.5, 126.5 – 135.5, ..., 171.5 – 180.5.

$\frac{n}{2} = 20$, so the median class is 144.5 – 153.5, with $l = 144.5$, $cf = 17$, $f = 12$, $h = 9$.

Median $= 144.5 + \frac{20 - 17}{12} \times 9 = 144.5 + 2.25 = 146.75$.

Answer: 146.75 mm

Q5 The lifetimes of 400 neon lamps are given below. Find the median lifetime.

Lifetime (hours)	1500 – 2000	2000 – 2500	2500 – 3000	3000 – 3500	3500 – 4000	4000 – 4500	4500 – 5000
Number of lamps	14	56	60	86	74	62	48
Cumulative frequency	14	70	130	216	290	352	400

SOLUTION

$\frac{n}{2} = 200$, so the median class is 3000 – 3500, with $l = 3000$, $cf = 130$, $f = 86$, $h = 500$.

Median $= 3000 + \frac{200 - 130}{86} \times 500 \approx 3406.98$.

Answer: about 3406.98 hours

Q6 The numbers of letters in 100 surnames are given below. Find the median, the mean and the modal size.

Number of letters	1 – 4	4 – 7	7 – 10	10 – 13	13 – 16	16 – 19
Number of surnames	6	30	40	16	4	4
Cumulative frequency	6	36	76	92	96	100

SOLUTION

Median: $\frac{n}{2} = 50$, so the median class is 7 – 10, with $l = 7$, $cf = 36$, $f = 40$, $h = 3$. Median $= 7 + \frac{50 - 36}{40} \times 3 = 8.05$.

Mean: the class marks are 2.5, 5.5, 8.5, 11.5, 14.5, 17.5, so $\sum f_i x_i = 15 + 165 + 340 + 184 + 58 + 70 = 832$. Mean $= \frac{832}{100} = 8.32$.

Mode: the modal class is 7 – 10, with $f_1 = 40$, $f_0 = 30$, $f_2 = 16$. Mode $= 7 + \frac{10}{80 - 46} \times 3 \approx 7.88$.

Answer: median 8.05, mean 8.32, mode ≈ 7.88 letters

MathSetu

Q7 The weights of 30 students are given below. Find the median weight.

Weight (kg)	40 – 45	45 – 50	50 – 55	55 – 60	60 – 65	65 – 70	70 – 75
Number of students	2	3	8	6	6	3	2
Cumulative frequency	2	5	13	19	25	28	30

SOLUTION

$\frac{n}{2} = 15$, so the median class is 55 – 60, with $l = 55$, $cf = 13$, $f = 6$, $h = 5$.

Median $= 55 + \frac{15 - 13}{6} \times 5 \approx 56.67$.

Answer: about 56.67 kg

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